

True/False – Balanced Conceptual Questions (15)

Instructions: 50% of these statements are True and 50% are False. Read carefully and reason your choice. Answers are included at the end.

- 1) Robotics involves the design, construction, and operation of machines capable of performing tasks automatically.
- 2) Unmanned surface vehicles (USVs) operate below the sea surface to perform underwater research.
- 3) The term 'actuator' refers to the computational core of a robot that processes sensor data.
- 4) The Sense–Plan–Act model describes the fundamental operational sequence in most robotic systems.
- 5) Hydraulic actuators are often used when high force and power are required.
- 6) Localization helps a robot determine its exact position in an environment.
- 7) The main advantage of hierarchical control is that it allows real-time, fast decision-making at the lowest level.
- 8) IoT integration in robotics enables devices to communicate and share data intelligently over networks.
- 9) Asimov's Laws of Robotics emphasize machine safety over human well-being.
- 10) A gripper-type end effector is specifically designed to resemble the human hand in structure and function.
- 11) In the hybrid control paradigm, planning and reactive layers coexist to achieve flexible and adaptive control.
- 12) The Four D's of Robotics (Dirty, Dull, Dangerous, Difficult) describe human-friendly tasks suited for manual labor.
- 13) Electric actuators provide smoother and cleaner operation compared to hydraulic systems.
- 14) A microcontroller is better suited for high-level planning and data visualization than a single-board computer.
- 15) Robots in the Fourth Industrial Revolution are increasingly interconnected and capable of autonomous decision-making.

Answer Key

- 1) True
- 2) False – USVs operate on the surface of the water, not underwater.
- 3) False – An actuator converts control signals into physical motion.
- 4) True
- 5) True
- 6) True
- 7) False – Hierarchical systems are slower due to planning steps.
- 8) True
- 9) False – Asimov's laws prioritize human safety above all.
- 10) False – The anthropomorphic type resembles the human hand, not a simple gripper.
- 11) True
- 12) False – The Four D's identify tasks *unsuitable* for humans, ideal for robots.
- 13) True
- 14) False – Microcontrollers handle low-level control, not high-level visualization.
- 15) True